

Cross-Validation Equilibrium

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Motivation

Use of machine learning to form predictive beliefs has become ubiquitous

- default risks in lending
- pretrial risk assessments
- likelihood of developing a disease

Decision making is not automatic, but remains in human control

- banker decides on credit terms
- judge sets bail
- doctors decide on treatment

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Question

What are the effects of predictive machine learning when players use it in a static game with incomplete information?

Contribution and Overview

- Study strategic interactions with the following features:
 - players task ML agent with choosing predictive models
 - ML agent sees noisy training data about payoff relevant outcome
 - predictive model yields belief over payoff relevant outcomes
 - players best-respond to these beliefs

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- Main interest: two-way linkage b/w player behaviour and model selection

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 - players best-respond to these beliefs
- Crucially, training data is drawn from endogenous distribution generated by other players' behaviour
- Main interest: two-way linkage b/w player behaviour and model selection
- We propose an equilibrium concept that captures these features:
Cross-Validation Equilibrium
- Focus on a core technique of machine learning: cross-validation

Road Map

- Related Literature
- Setting
- Cross-Validation Equilibrium
- Results
- Applications

Related Literature

Related Literature

- ML in strategic interactions:
Calvano et al. (2020), Hansen et al. (2021), Asker et al. (2024), Banchio and Mantegazza (2024), Possnig (2024), Cartea et al. (2025), Waizmann (2025), ...
model selection and prediction instead of reinforcement learning
- ML for belief formation in games:
Jehiel and Weber (2025), Spiegler (2026)
cross-validation as model selection criterion
- Equilibrium concepts with non-rational expectations:
Jehiel (2005), Eyster and Rabin (2005), Esponda (2008), Esponda and Pouzo (2016), Spiegler (2016); Osborne and Rubinstein (1998), Salant and Cherry (2020), Danenberg and Spiegler (2025)
add ML based model selection

Model and Equilibrium Concept

Setting

Static Bayesian game:

- n players
- types T_i
- actions A_i
- states of nature Θ
- prior μ over $\Theta \times T_i$
- payoff-relevant outcome $x_i \in \mathbb{R}$

$$x_i = f_i(\theta, a_{-i})$$

- utility function is linear in x_i

$$u_i(a_i, t_i, x_i)$$

Notation

- Strategy for player i is a map

$$\sigma_i : T_i \rightarrow \Delta(A_i)$$

- Strategy profile σ induces distribution

$$p^\sigma(\theta, t, a) = \mu(\theta, t) \cdot \prod_{i \in N} \sigma_i(a_i \mid t_i)$$

- Expected value of sufficient statistic given type t_i is

$$\bar{x}_i^\sigma(t_i) = \sum_{\theta, a_{-i}} p^\sigma(\theta, a_{-i} \mid t_i) f_i(\theta, a_{-i})$$

ML Agent

- Each player i has access to a *machine learning agent*
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- Given σ , ML agent has a *training sample*: for each type t_i

$$\bar{x}_i^\sigma(t_i) + \varepsilon_{t_i}$$

- $\varepsilon_{t_i} \sim v(\cdot)$ symmetric distribution around 0 with finite support
 - iid across types and players
- Given σ , ML agent has a *validation sample*: for each type t_i

$$\bar{x}_i^\sigma(t_i) + \eta_{t_i}$$

- η_{t_i} has mean 0
 - iid across types and players

Model Selection

- A *predictive model* for player i is a partition Π_i of T_i
- Set of feasible models for player i is $\mathcal{P}_i$

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- For $M \subset T_i$, denote

$$\hat{x}^{\varepsilon_i}(M) = \sum_{t_i \in M} \mu(t_i \mid t_i \in M)(\bar{x}_i^\sigma(t_i) + \varepsilon_{t_i})$$

- ML agent selects model based on out-of-sample prediction error given training and validation sample

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- ML agent selects model based on out-of-sample prediction error given training and validation sample
- Model Π minimizes out-of-sample prediction error if

$$\Pi_i \in \arg \min_{\Pi'_i \in \mathcal{P}_i} \mathbb{E}_{\eta} \sum_{t_i \in T_i} \mu(t_i) [\hat{x}^{\varepsilon_i}(\pi'(t_i)) - (\bar{x}_i^{\sigma}(t_i) + \eta_{t_i})]^2$$

Beliefs and Behaviour

- Selected model and training sample realization determine players' beliefs about payoff-relevant outcome
- Given model Π_i and training sample realization $(\bar{x}_i^\sigma + \varepsilon_{t_i})_{t_i}$, player i of type t'_i 's belief is

$$\hat{x}^{\varepsilon_i}(\pi_i(t'_i)) = \sum_{t_i \in \pi_i(t'_i)} \mu(t_i \mid t_i \in \pi_i(t'_i)) (\bar{x}_i^\sigma(t_i) + \varepsilon_{t_i})$$

- Type t'_i of player i best-responds to this belief by choosing

$$a'_i \in \arg \max_{a_i \in A_i} u_i(a_i, t_i, \hat{x}^{\varepsilon_i}(\pi_i(t'_i)))$$

Cross-Validation Equilibrium

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Compatibility

A pair (Π_i, g_i) is compatible with σ_{-i} and ε_i if

- partition Π_i minimizes out-of-sample prediction error
- for every type t_i , $g_i(t_i)$ is a best-response to beliefs induced by $\pi_i(t_i)$

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Cross-Validation Equilibrium

Strategy profile σ is a *Cross-Validation Equilibrium* if for every player i and training sample realization ε_i , there is a mixture β_{ε_i} over pairs (Π_i, g_i) compatible with σ_{-i} and ε_i , such that

$$\sigma_i(a_i \mid t_i) = \sum_{\varepsilon_i} v(\varepsilon_i) \sum_{(\Pi_i, g_i) | g_i(t_i) = a_i} \beta_{\varepsilon_i}(\Pi_i, g_i) \quad \forall t_i \in T_i$$

An Example

		Player 2		Player 2	
		$a_2 = 0$	$a_2 = 1$	$a_2 = 0$	$a_2 = 1$
Player 1	$a_1 = 0$	0, 0	0, -1	0, 0	0, -1
	$a_1 = 1$	-1, 0	-1, -1	-1, 0	1/3, 1/3
		$\theta = 0$		$\theta = 1$	

- State θ commonly known and equally likely
- Let $f_i(\theta, a_j) = a_j$
- $\bar{x}^\sigma(\theta) = \sigma_j(1|\theta)$ is proba player j plays $a_j = 1$ in state θ

$$u_i(a_i, \theta, x) = \begin{cases} -a_i & \theta = 0 \\ a_i(\frac{4}{3}x - 1) & \theta = 1 \end{cases}$$

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- Assume $\varepsilon \sim \mathcal{U}\{-d, d\}$ with $d > \frac{1}{2}$
- In $\theta = 0$, $a_i = 0$ independent of $x_i \rightarrow \bar{x}(0) = 0$
- In $\theta = 1$, $a_1 = 1$ only if $\bar{x}(1) \geq 3/4$
- Prediction error from fine partition $\frac{1}{2}(\varepsilon_0^2 + \varepsilon_1^2) + \sigma_\eta^2$
- Prediction error from coarse partition $(\frac{\bar{x}_1}{2} + \frac{\varepsilon_0 + \varepsilon_1}{2})^2 + \sigma_\eta^2$
- There is a CVE with $\bar{x}(1) = 1/4$:
 - $\varepsilon_1 = -\varepsilon_0$: adopt coarse partition and play $a_i = 0$
 - $\varepsilon_1 = \varepsilon_0 = d$: adopt fine partition and play $a_i = 1$
 - $\varepsilon_1 = \varepsilon_0 = -d$: adopt fine partition and play $a_i = 0$

Results

Existence and Convergence

Assume the game is finite.

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Let $v^n(\cdot)$ be a sequence of finitely-supported distributions with $\text{var } v^n \rightarrow 0$, and σ^n a corresponding sequence of CVEs

Proposition: Convergence

Assume each player's set of feasible partitions contains a finest partition. If the limit $\sigma = \lim_{n \rightarrow \infty} \sigma^n$ exists, then σ is an ABEE with respect to each player's finest partition Π_i^{finest} .

definition ABEE

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Corollary

If each player has access to the partition $\{\{t_i\}_{t_i \in T_i}\}$, then $\lim \sigma^n$ is a Bayes-Nash equilibrium.

Lemmas

Lemma

Assume Π' refines Π . Suppose that σ_{-i} is such that $\bar{x}_i^\sigma(\cdot)$ is measurable with respect to Π . Then the expected out-of-sample prediction error from partition Π is weakly lower than from partition Π' for any realization of the training sample noise ε_i .

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Lemma

For every $t_i \in T_i$ and any partition $\Pi_i \in \mathcal{P}_i$, the expectation of ε_{t_i} conditional on the event that Π_i minimizes the prediction error is zero.

Applications

Cournot Competition

- 2 players and 2 equally likely states $0 < \theta_1 < \theta_2$
- state θ is common knowledge, $t_1 = t_2 = \theta$ almost surely
- payoffs $u_i(t_i, a_i, x_i) = (\theta - x_i - a_i)a_i$
- sufficient statistic $f(\theta, a_j) = a_j$
- training sample noise $\varepsilon_\theta \sim \mathcal{N}(0, \sigma^2)$
- all partitions are feasible

Cournot Competition

- best-response of player i and type θ is

$$\begin{cases} \theta/2 - 1/2(\bar{x}(\theta) + \varepsilon_{\theta}) & \text{if fine partition is chosen} \\ \theta/2 - 1/4(\bar{x}(\theta_1) + \bar{x}(\theta_2) + \varepsilon_{\theta_1} + \varepsilon_{\theta_2}) & \text{if coarse partition is chosen} \end{cases}$$

- let λ be probability that fine partition is chosen in symmetric CVE
- in symmetric CVE,

$$\bar{x}(\theta_i) = \theta/2 - \bar{x}(\theta_i)(1 + \lambda)/4 - \bar{x}(\theta_j)(1 - \lambda)/4$$

- the fine partition is adopted if $|\varepsilon_{\theta_2} - \varepsilon_{\theta_1}| \leq |\bar{x}(\theta_2) - \bar{x}(\theta_1)|$
- hence,

$$\lambda = \mathbb{P} \left[|\varepsilon_{\theta_2} - \varepsilon_{\theta_1}| \leq \frac{\theta_2 - \theta_1}{2 + \lambda} \right]$$

Cournot Competition

Result

- There is a unique symmetric CVE.
- Ex-ante expected quantities coincide with ex-ante quantities in Nash equilibrium.
- Equilibrium quantities are more volatile than in Nash equilibrium.

Team Effort

- Team of 2 players exert effort $a_i \in \mathbb{R}$ to affect project success
- Commonly observed state θ affects probability of success, $0 < \theta_1 < \theta_2$ equally likely
- Payoffs

$$u_i(a_i, a_j, \theta) = \delta\theta(a_i + a_j) + (1 - \delta)a_i a_j - \frac{1}{2}a_i^2$$

for $\delta \in (0, 1)$

- Sufficient statistic $x_i = a_j$ equals effort of other player
- Unique Nash equilibrium $a_1^*(\theta) = a_2^*(\theta) = \theta$

Team Effort

- Training sample noise $\varepsilon_\theta \sim \mathcal{N}(0, \sigma^2)$
- In symmetric CVE, fine partition is adopted with probability

$$\lambda = \mathbb{P} \left(|\Delta\varepsilon| < \frac{\delta\Delta\theta}{1 - \lambda(1 - \delta)} \right) = 2\Phi \left(\frac{\delta\Delta\theta}{\sqrt{2}\sigma_\varepsilon[1 - \lambda(1 - \delta)]} \right) - 1$$

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Results

- Multiple CVE can exist.
- As $\delta \rightarrow 0$, $\lambda \rightarrow 0$ and $\bar{x}(\theta) \rightarrow 1/2(\theta_2 + \theta_1)$
- As $\delta \rightarrow 1$, $\lambda \rightarrow 1$ and $\bar{x}(\theta) \rightarrow \theta$
- Expected payoffs in CVE are always lower than in Nash equilibrium.

Jury Voting

- defendant is innocent $\theta = 0$ or guilty $\theta = 1$, both equally likely
- n voters who observe private, conditional independent signals t_i with $\mathbb{P}[t_i = \theta | \theta] = q \in (1/2, 1)$
- voters can acquit $a_i = 0$ or convict $a_i = 1$
- voting rule: unanimity
- voters receive payoff of 1 for correct decision, 0 otherwise

Jury Voting

- sufficient statistic: difference in payoff from convicting and acquitting

$$f(\theta, a_{-i}) = \mathbf{1}[\theta = 1 \cdot \prod_{j \neq i} a_j] - \mathbf{1}[\theta = 0 \cdot \prod_{j \neq i} a_j]$$

and $u_i(a_i, x_i) = (2a_i - 1)x_i$

- training sample noise ε has full support on $[-1, 1]$

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Result

In any symmetric CVE, as the number n of voters grows

- coarse partition gets selected with probability 1
- voters convict and acquit with probability $1/2$ independent of signal

details

Conclusion

- Study games when players delegate belief formation to predictive ML systems trained on endogenous, noisy data
- Cross-Validation Equilibrium captures a simple idea: players best-respond to predictions generated by models selected for out-of-sample performance
- Random samples lead to random model choice and stochastic behaviour
- Other applications:
 - speculative bets
 - illusion of control
- Next steps:
 - apply idea behind CVE to competitive selection markets
 - suggestions?

Thank you!

Definition: ABEE (Jehiel, 2005)

Definition

σ is an *Analogy Based Expectations Equilibrium* with respect to the partitions $\{\Pi_i\}$ if for each player i and type t_i , for all a_i with $\sigma_i(a_i|t_i) > 0$

$$\begin{aligned} & u_i \left(a_i, t_i, \sum_{t'_i \in \pi(t_i)} \mu(t'_i|t'_i \in \pi(t_i)) \bar{x}_i^\sigma(t'_i) \right) \\ & \geq u_i \left(a'_i, t_i, \sum_{t'_i \in \pi(t_i)} \mu(t'_i|t'_i \in \pi(t_i)) \bar{x}_i^\sigma(t'_i) \right) \end{aligned}$$

holds for all $a'_i \in A_i$.

Jury Voting

- let $A = q\sigma(1) + (1 - q)\sigma(0)$, $B = q\sigma(0) + (1 - q)\sigma(1)$
- then $\bar{x}(1) = qA^{n-1} - (1 - q)B^{n-1}$ and $\bar{x}(0) = (1 - q)A^{n-1} - qB^{n-1}$
- so that $\Delta\bar{x} = \bar{x}(1) - \bar{x}(0) = (2q - 1)(A^{n-1} + B^{n-1})$
- when adopting the fine partition, choose $a_i = 1$ iff $\bar{x}(t) + \varepsilon_t > 0$
- when adopting coarse partition, choose $a_i = 1$ iff $\frac{\bar{x}(1) + \varepsilon_1 + \bar{x}(0) + \varepsilon_0}{2} > 0$
- $\sigma(t)$ bounded away from 1 independent of number of voters
- sufficient statistic $\bar{x}(t) \rightarrow 0$ as number of voters grows
- coarse partition is adopted with probability $\rightarrow 1$
- voting becomes unresponsive to signal

Application: Speculative Bets